Public Transport Scheduling.

Problem statement
optimize bus departure time and vehicle assignment

objective :-
Reduce passenger waiting time and operation cost

Pseudocode

BEGIN

1. Define trips, passenger arrival timings, available, departure time windows and operation cost
2. Generate a initial population with random schedule
3. Evaluate fitness of each schedule
   a. calculate total passenger waiting time
   b. calculate total operating cost
   c. Add penalties for vehicle conflicts and invalid schedule
   d. calculate total objective value
4. while Max No: of generation is not reached

   do

   a. select best schedules as parents
   b. Apply crossover to combine depart -ure timings and vehicle assignme
   c. Apply Mutation to change depar -ure times
   d. Repair invalid schedules if Necessary
   e. evaluate fitness of the new

schedule
8. Replace weaker schedules with better schedules

END WHILE

5. Return the schedule with the lowest objective value:
   * optimized bus departure times
   * vehicle assignments
   * passenger waiting time
   * operating cost

END

output

OPTIMIZED BUS SCHEDULE

Trip 1: Departure = 16 min, vehicle = 1, waiting = 18 passeng-Min, cost = 200

Trip 2: Departure = 31 min, vehicle = 2, Waiting = 18 passenger-Min, cost = 300

Trip 3: Departure = 46 min, vehicle = 1, waiting = 18 passenger-Min, cost = 200

Trip 4: Departure = 61 Min, vehicle = 2, Waiting = 18 passenger-Min, cost = 300

Total passenger waiting time: 72 passenger-minutes

Total operating cost: 1000

objective value: 272.0

o/p seen getting value